

Notes (Week 4 Wednesday)

These are the notes I wrote during the lecture.

Well-formed formulas:

the syntactically correct propositional
logic formulas

Valuations, boolean algebras

The boolean algebra of truth values
 $\{\text{true}, \text{false}\} \in \mathbb{B}$

Given a *valuation* v :

a function that assigns truth values
to variables (aka propositional atoms)

We can define a function $\llbracket \cdot \rrbracket v$

which given a valuation maps formulas
to truth values

$$\llbracket p \wedge q \rrbracket (p \mapsto \text{true}, q \mapsto \text{false}) = \text{false}$$

This gives us enough tools to give
fully formal meaning to $\equiv$ (logical equivalence)

$$\phi \equiv \psi \quad \text{iff for every } v: \llbracket \phi \rrbracket v = \llbracket \psi \rrbracket v$$

Equivalently: the truth tables of ϕ and ψ
are the same

$\Rightarrow$ logical entailment

$\phi \Rightarrow \psi$ iff for every v
whenever $\llbracket \phi \rrbracket v = \text{true}$
then $\llbracket \psi \rrbracket v = \text{true}$

From now on, we'll switch notation for entailment
and its type

$$\Rightarrow : \text{wff} \times \text{wff} \rightarrow \mathbb{B}$$

$\models$ "entails" or "models"

$T \models \phi$ ϕ is still a wff
 T is a *theory*

a theory is a set of wffs

$T \models \phi$ iff
 for every valuation v such that
 if for every $\psi \in T$, $\llbracket \psi \rrbracket v = \text{true}$
 then $\llbracket \phi \rrbracket v = \text{true}$

$$p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

A formula is in CNF if:
 it is a conjunction of
 disjunctions of
 atoms or negated atoms

Some more formulas:

$\perp$ <-- a unary conjunction of a nullary
 disjunction

$\top$ <-- counts as a 0-ary "nullary" conjunction
 the conjunction of 0 formulas

a <-- this also counts!

$a \vee b$ <-- yes this counts!
 we can think of it as a conjunction
 of a single formula ($a \vee b$)

$a \wedge (b \vee (a \wedge b))$
 ^^^^^ that's a conjunction
 which is neither an
 atom nor a negated atom

Is this formula satisfiable?

$$(p \wedge q \wedge r \wedge f) \vee (q \wedge f \wedge g \wedge \neg f) \\ \vee (\neg r \wedge g \wedge b \wedge c \wedge \neg g \wedge s)$$

When the formula is in DNF
 to satisfy the formula, we only need
 to satisfy one disjunct
 A conjunction of literals is satisfiable
 iff it has no contradictory literals

CNF formula.

With CNF, we can easily check validity.

$$(p \vee q \vee r \vee f) \wedge (q \vee f \vee g \vee \neg f) \\ \wedge (\neg r \vee g \vee b \vee c \vee \neg g \vee s)$$

Does **every** conjunct have contradictory literals?

If yes, it's valid. If no, it's not.

$$\neg\neg\phi \equiv \phi$$

$$\neg(\phi \wedge \psi) \equiv \neg\phi \vee \neg\psi$$

natural deduction is a syntactic proof technique because we never:

- construct truth tables
- consider valuation functions ever
- we never use these weird brackets []

Inference rules are crafting recipes

If I have nine iron ingots, I can make
iron block

iron	iron	iron	iron	iron	iron	iron	iron	iron
iron block								

^ iron block-introduction

iron block
9 x iron ingots

^ iron block-elimination

Why disassemble the block

A: You might only want an ingot.

e.g. pickaxe for three ingots, two sticks

Every operator comes with:

- introduction rules for proving things with that operator outermost
- elimination rules for proving **consequences of** statements with that operator outermost

We write

$T \vdash \varphi$ means " φ is provable using ND rules from T "

These rules are sound and complete.

$T \vdash \varphi$ iff $T \models \varphi$